

Converting Casual Riders to Annual Members at Capital Bikeshare

Data-Driven Station Targeting Strategy

Five high yield stations identified in the upper Northwest DC corridor. A validated pilot scales to an estimated 1,060 new annual members and an estimated \$127,000 ARR systemwide

The Business Question

Capital Bikeshare leaves recurring revenue on the table: around 30% of rides are casual, behave like members

Situation

Capital Bikeshare operates over 6 million trips per year across the DC region. Annual members generate predictable recurring revenue; casual riders pay per ride and represent around 30% of trips, a substantial pool of repeat behavior that converts only sporadically



Complication

Without user IDs in the public dataset, traditional conversion analysis is blocked. The dataset records rides, not riders, meaning the unit of observation is the trip, not the customer



Question

Which 3–5 stations represent the highest ROI targets for a casual to member conversion campaign?



Analytical Pivot

Analysis operates at the ride level via behavioral signatures, not individual user tracking. Rides with temporal and trip patterns resemble member ridership flag the conversion opportunity even without seeing the rider

The Approach

A gradient boosting classifier learns the behavioral identity of member ridership and scores every casual ride against it

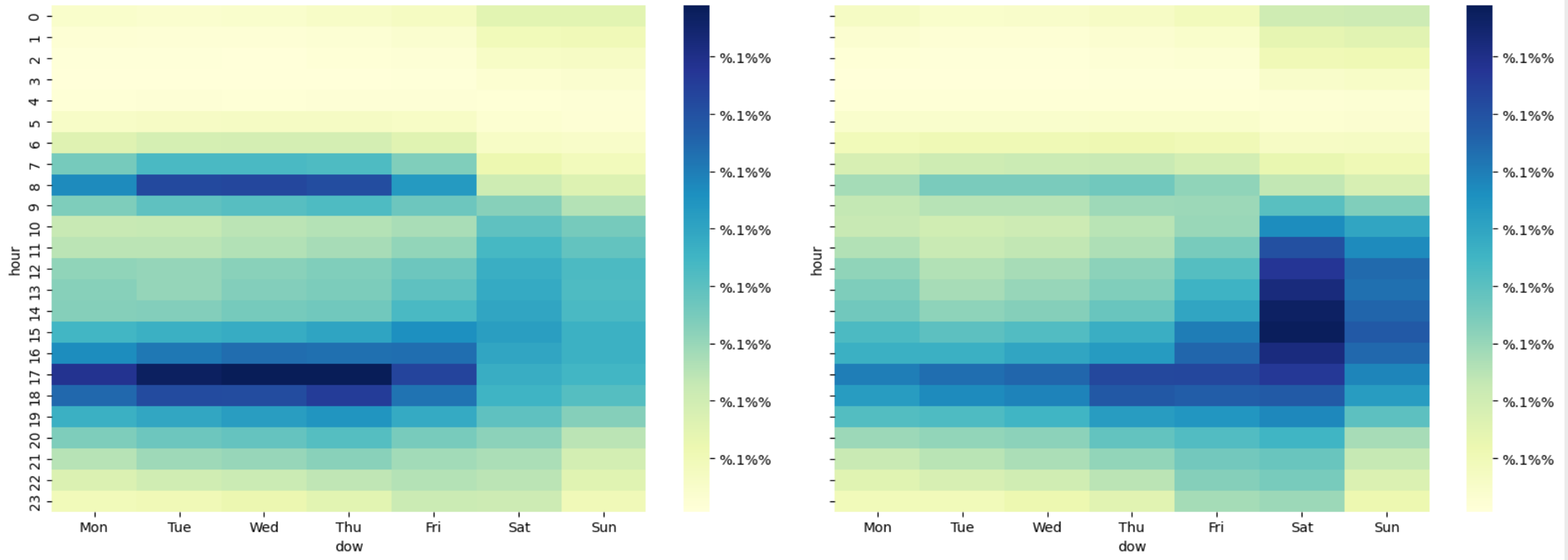
1. **Data:** 6.2M rides, 12 months (May 2025 – Apr 2026), 500 stations
2. **Cleaning:** 2.3% rides dropped (duration outliers); systematic data quality audit
3. **Features:** 17 engineered features: temporal, trip-level, station aggregates
4. **Modeling:** chronological train/val/test split, HistGradientBoostingClassifier
5. **Outcome:** ROC AUC = 0.73 on validation; Cramér's V = 0.21 for rush-hour effect

Note: AUC of 0.73 reflects real overlap between member and casual ride patterns. Effect sizes confirm meaningful behavioral differences; sample size makes p-values uninformative.

The Central Pattern

Members typically ride in commute rhythm. Casuals typically ride in leisure rhythm, but are overlapping patterns

Figure 3.9: Heatmap of Ride Counts by Hour and Day of Week for Members vs Casuals



Member panel: double-hump at 7–9am and 4–6pm, weekdays.

Casual panel: Saturday and Sunday dominant. The textbook leisure pattern with a mid day increase during the week. The visible overlap during weekday rush hours is the conversion opportunity.

The Finding

55,000 casual rides per month look like member rides, 5 times the size of a simple commuter heuristic

170,968

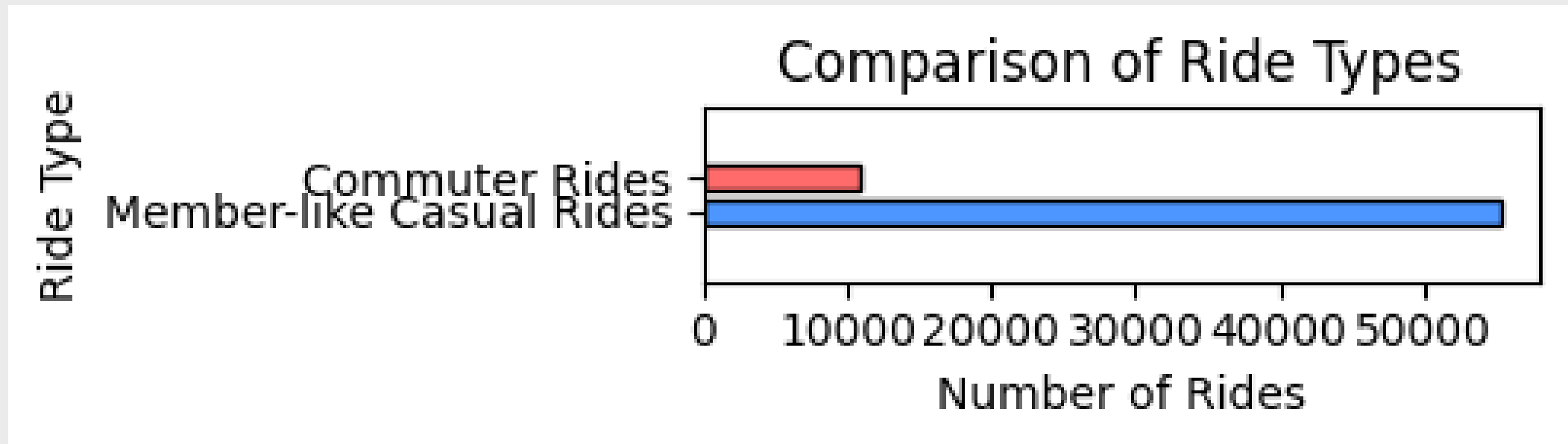
total casual rides per month

55,226 (32%)

Model flagged as behaviorally
member-like

5×

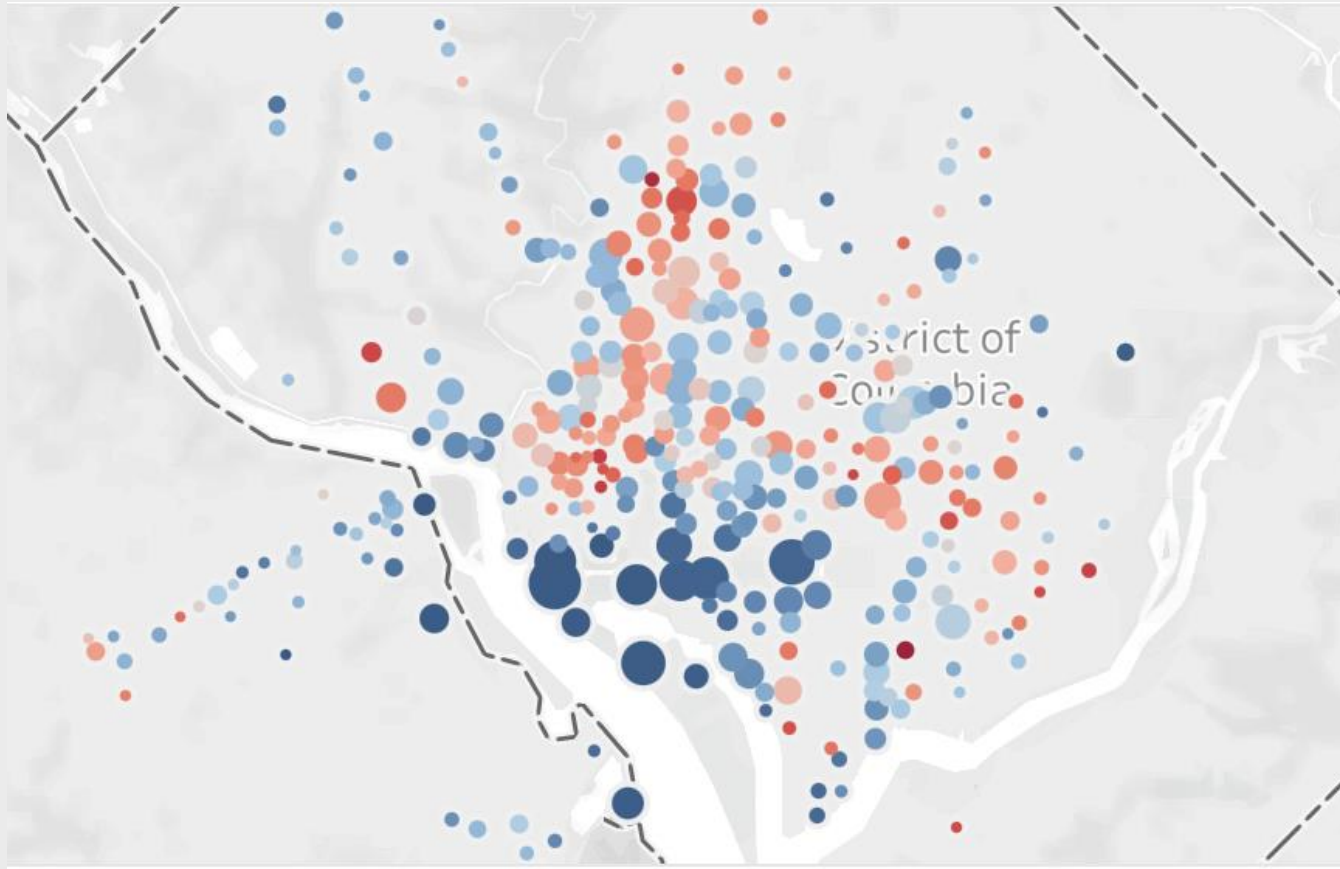
the model's reach vs a rule
based commuter heuristic



These are riders who already behave like members. They just haven't purchased the membership.

The Targets and Opportunity

Five stations in the upper Northwest DC corridor concentrated on the conversion opportunity



Station	Monthly member-like rides	Concentration	Total casual rides
Columbus Circle / Union Station	528	44%	1,206
14th & Irving St NW	490	59%	826
New Hampshire Ave & T St NW	465	43%	1,078
37th & O St NW / Georgetown	407	51%	798
14th & V St NW	380	40%	955

The Recommendation

Pilot a four-tactic corridor campaign; scale once validated

Interventions

1. **On bike screen offers during weekday rush hours.** Around 1,400 impressions per station per month at zero incremental cost
2. **Permanent dock signage with QR value calculator.** "Save \$270/year" messaging at all five docks
3. **Trigger based email/push to repeat single trip buyers.** Operationalizes the same insight via transaction data
4. **Adjacent employer and university partnerships.** Anchor: Georgetown University, Union Station's federal cluster

Sizing

Pilot (5 stations) 43 new annual members around **\$5,200** ARR

Scaled systemwide 1,060 new annual members **\$127K** ARR
*Sensitivity: **\$48K–\$318K** depending on conversion rate (3%–20%).*

Pilot Design, Limitations, Next Steps

A 90 day controlled pilot validates the mechanic before systemwide rollout.

Pilot Design

- 5 pilot stations vs 3 matched control stations in 90 day window
- Success metric: $\geq 20\%$ lift in casual to member conversion rate at $p < 0.05$. 25 conversions per arm required

Limitations

- No user IDs available
- Analysis is ride level AUC 0.73 is moderate; field validation required
- Top features encode geographic base rate, not pure behavior

What I'd do next

- Acquire anonymized user level data
- Add weather, event, and station amenity features
- Hyperparameter tuning and out of time validation
- Operationalize the model in a real time targeting system